

On the other hand, the fundamental group of a wedge sum of countably many circles is countably generated, hence countable.

The group $\pi_1(X)$ is actually far more complicated than $\prod_{\infty} \mathbb{Z}$. For one thing, it is nonabelian, since the retraction $X \rightarrow C_1 \cup \cdots \cup C_n$ that collapses all the circles smaller than C_n to the basepoint induces a surjection from $\pi_1(X)$ to a free group on n generators. For a complete description of $\pi_1(X)$ see [Cannon & Conner 2000].

It is a theorem of [Shelah 1988] that for a path-connected, locally path-connected compact metric space X , $\pi_1(X)$ is either finitely generated or uncountable.

Applications to Cell Complexes

For the remainder of this section we shall be interested in 2-dimensional cell complexes, analyzing how the fundamental group is affected by attaching 2-cells. According to an exercise at the end of this section, attaching cells of higher dimension has no effect on π_1 , so all the interest lies in how the 2-cells are attached.

Suppose we attach a collection of 2-cells e_{α}^2 to a path-connected space X via maps $\varphi_{\alpha}: S^1 \rightarrow X$, producing a space Y . If s_0 is a basepoint of S^1 then φ_{α} determines a loop at $\varphi_{\alpha}(s_0)$ that we shall call φ_{α} , even though technically loops are maps $I \rightarrow X$ rather than $S^1 \rightarrow X$. For different α 's the basepoints $\varphi_{\alpha}(s_0)$ of these loops φ_{α} may not all coincide. To remedy this, choose a basepoint $x_0 \in X$ and a path γ_{α} in X from x_0 to $\varphi_{\alpha}(s_0)$ for each α . Then $\gamma_{\alpha}\varphi_{\alpha}\bar{\gamma}_{\alpha}$ is a loop at x_0 . This loop may not be nullhomotopic in X , but it will certainly be nullhomotopic after the cell e_{α}^2 is attached. Thus the normal subgroup $N \subset \pi_1(X, x_0)$ generated by all the loops $\gamma_{\alpha}\varphi_{\alpha}\bar{\gamma}_{\alpha}$ for varying α lies in the kernel of the map $\pi_1(X, x_0) \rightarrow \pi_1(Y, x_0)$ induced by the inclusion $X \hookrightarrow Y$.

Proposition 1.26. *The inclusion $X \hookrightarrow Y$ induces a surjection $\pi_1(X, x_0) \rightarrow \pi_1(Y, x_0)$ whose kernel is N . Thus $\pi_1(Y) \approx \pi_1(X)/N$.*

It follows that N is independent of the choice of the paths γ_{α} , but this can also be seen directly: If we replace γ_{α} by another path η_{α} having the same endpoints, then $\gamma_{\alpha}\varphi_{\alpha}\bar{\gamma}_{\alpha}$ changes to $\eta_{\alpha}\varphi_{\alpha}\bar{\eta}_{\alpha} = (\eta_{\alpha}\bar{\gamma}_{\alpha})\gamma_{\alpha}\varphi_{\alpha}\bar{\gamma}_{\alpha}(\gamma_{\alpha}\bar{\eta}_{\alpha})$, so $\gamma_{\alpha}\varphi_{\alpha}\bar{\gamma}_{\alpha}$ and $\eta_{\alpha}\varphi_{\alpha}\bar{\eta}_{\alpha}$ define conjugate elements of $\pi_1(X, x_0)$.

Proof: Let us expand Y to a slightly larger space Z that deformation retracts onto Y and is more convenient for applying van Kampen's theorem. The space Z is obtained from Y by attaching rectangular strips $S_{\alpha} = I \times I$, with the lower edge $I \times \{0\}$ attached along γ_{α} , the right edge $\{1\} \times I$ attached along an arc in e_{α}^2 , and all the left edges $\{0\} \times I$ of the different strips identified together. The top edges of the strips are not attached to anything, and this allows us to deformation retract Z onto Y .

